

FINAL EXPLANATION: MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: MATHEMATICS GRADE 10 | SUB-STRAND 1.3: QUADRATIC EQUATIONS | FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Final Explanation: Planning the Githurai Harambee Hall. Driving question: How can we use quadratic equations to find unknown dimensions and quantities that help our community solve real planning problems, like building the Harambee hall in Githurai? Use what you observed and learned in Lessons 1 to 7 to answer each question in the space under it. Show every step of your working, write in full sentences when you explain, and always say what your answer means for the community. Use the Githurai hall data: the floor is a rectangle with an area of 120 m^2 , the width is w and the length is $w + 2$. You may use a scientific calculator for Section 4.

Section 1: Why a linear equation is not enough

The Githurai committee wants a rectangular hall floor of area 120 m^2 , with the length 2 m more than the width w . (a) Write an equation for the area in terms of w . (b) Expand it and write it in the form $ax^2 + bx + c = 0$. State the values of a , b and c . (c) Explain why trial-and-error and linear equation methods did not work cleanly for the committee. How is this equation different from a linear equation?

Write your answer here:

Section 2: Solving the hall problem by factorisation

(a) Factorise $w^2 + 2w - 120 = 0$. Explain how you chose your two numbers. (b) Use the Zero Product Property to find the values of w . (c) Which value will the committee use for the width, and why? State the length of the hall. (d) Check that your dimensions give an area of 120 m^2 .

Write your answer here:

Section 3: Three methods, one hall

(a) Show how you would solve $w^2 + 2w - 120 = 0$ by completing the square. (b) Show how you would solve it using the quadratic formula. (c) Compare the three methods you have learned (factorisation, completing the square and the quadratic formula). When is each most useful? Use the Nakuru school garden equation $x^2 + 6x = 48$ or another example from class to support your answer.

Write your answer here:

Section 4: Interpreting solutions in another community context

In Lesson 6, a stone was thrown upward from a Mombasa cliff, with height $h(t) = -5t^2 + 20t + 25$ metres after t seconds. (a) Write the equation you must solve to find when the stone hits the sea. (b) Use the discriminant to say how many real solutions there will be, before you solve. (c) Solve the equation and state which solution you accept. Explain your choice in a Claim-Evidence-Reasoning format.

Write your answer here:

Section 5: Answering the driving question

Write a clear explanation that answers the driving question: How can we use quadratic equations to find unknown dimensions and quantities that help our community solve real planning problems, like building the Harambee hall in Githurai? Use the following in your answer: (i) how quadratic expressions and equations differ from linear ones, (ii) the steps of your unit journey from the hall sketch to a solution, and (iii) why interpreting solutions in context matters. Include at least one other community example from the unit.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Modelling the situation as a quadratic equation	Writes $w(w + 2) = 120$ correctly, expands and rearranges accurately to $w^2 + 2w - 120 = 0$, identifies a, b and c, and explains clearly why the equation is not linear.	Writes and expands the equation correctly and identifies a, b and c, with a partial explanation of why it is not linear.	Has errors in forming or expanding the equation, or cannot identify a, b and c or explain how it differs from a linear equation.
Accuracy of solution methods	Solves correctly by factorisation, completing the square and the quadratic formula, with every step shown and checked, and uses the discriminant correctly.	Solves correctly with at least two methods and shows most steps, with minor slips in the other method or the discriminant.	Makes major errors in methods or steps, or uses only one method with limited accuracy.
Interpreting solutions in context	Rejects negative roots with clear reasoning, states hall dimensions of 10 m by 12 m with units, verifies the area of 120 m ² , and applies the same reasoning to the Mombasa problem.	Chooses the correct root and states the dimensions with a brief reason, with some missing checks or units.	Accepts both roots without interpretation or gives no reason for the choice.
Comparing methods and using evidence	Explains when each method is most useful with specific examples from the unit, such as the Nakuru garden, and uses Claim-Evidence-Reasoning.	Describes when each method is useful, with general or limited examples.	Gives vague or incorrect comparisons with no supporting examples.

Communicating the answer to the driving question	Gives a coherent, well-organised explanation linking the unit journey to community planning, with precise mathematical vocabulary and another Kenyan example.	Gives a mostly clear explanation of the journey and its community use, with some gaps in vocabulary or links.	Gives a disjointed or incomplete explanation that does not link quadratics to community problems.
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ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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